Centre for Early Mathematics Learning

Mathematical Explanations in the Classroom

Self-explanation improves learning

Research has shown that encouraging learners to produce their own explanations in mathematics is an effective strategy for improving their learning. This is called ***self-explanation***. Research shows that self-explanation increases student learning of new math content, such as worked examples, or proofs. A summary of findings from 44 different studies found that prompting self-explanations led to sizable increases in learning, compared to when teachers didn't use self-explanation (Rittle-Johnson et al., 2017).

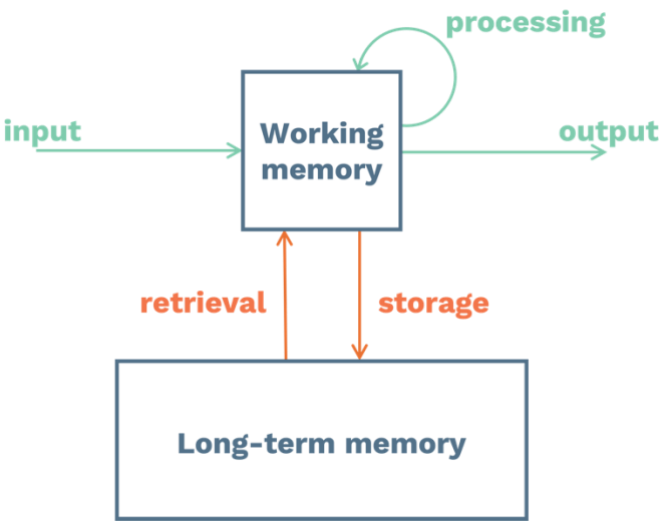
Self-explanations are statements generated by the learner that makes clear why some mathematical statement is true. To benefit learning, self-explanations should draw on learners' prior knowledge to explain why some mathematical statement is true.

How self-explanation improves learning

Self-explanation is thought to promote learning via knowledge integration. Self-explanation supports knowledge integration because it requires learners to activate relevant prior knowledge and connect it with new information. This new knowledge will then be integrated with, and stored alongside, the learner's existing knowledge.

The diagram below shows a simple cognitive model, depicting how researchers think we process incoming information.

Working memory is where we process new information. However, we can't store information in working memory for very long. Instead, all our prior knowledge is stored in our **long-term memory**. Securely learning something new requires integrating the new learning into long-term memory.



The **green arrows** show some new information entering working memory (input), being processed in working memory (processing), before potentially leaving working memory (output).

The **orange arrows** represent thinking processes that can be activated by prompting a learner to self-explain the information in working memory:

1. **retrieving** existing knowledge from long-term memory to integrate it with the new information in working memory, in order to provide an explanation of the new information
2. **storing** the new knowledge alongside the prior knowledge, back into long-term memory

In sum, self-explanation supports learning by prompting the learner to recall prior knowledge and integrate this with new knowledge, which helps store the new knowledge alongside the prior knowledge in long term memory.

Using self-explanation in teaching

Ideally, explanations should be generated by learners, rather than by the teacher. This is because it gives learners the opportunity to actively retrieve prior knowledge and integrate it with the new knowledge. Teachers can support learners to do this by using ***self-explanation prompts***, such as asking them why they believe a mathematical statement to be true. A good way to get learners to practice self-explanation is to ask them to explain in front of peers.

For example, a teacher could ask learners to calculate $\frac{1}{4} + \frac{2}{5}$. After the learners have given the correct answer of $\frac{13}{20}$ or explained how they calculated the answer, the teacher could then prompt the learners to explain why that is the correct answer: "why does that method give the correct answer?". This prompts the learner to consider and identify the prior knowledge that they used to calculate the answer. This will allow them to integrate the maths in question with their prior knowledge, and develop their understanding for similar questions.

The learner might then give the following explanation:

"When adding fractions, the denominators need to have the same value, so you need to find the lowest common multiple of the denominators."

The learner has explained why they multiplied the numerator and denominator of each fraction by five and four respectively, not just stated the strategy that they used. Notice how this required the learner to recall some prior knowledge: when adding fractions, they need to have the same denominator, and the easiest choice is the lowest common multiple of both denominators. By activating this prior knowledge, it increases the chance that the new knowledge will be stored in long-term memory alongside the prior knowledge, and that learners will be more confident when faced with another fraction sum.

Sometimes learners give responses that describe how they got their answer, but fall short of providing explanation. For example, they might say:

"I multiplied the numerator and denominator of one quarter by 5, and multiplied the numerator and denominator of two fifths by 4, which gave five twentieths and eight twentieths which equals thirteen twentieths."

The learner has reported on the strategy that they used to answer the question rather than explaining why they used this strategy, or why the strategy gives the correct answer. The learner has not provided any evidence that they have retrieved prior knowledge that they used to answer the question. The teacher cannot verify from the learner's response that they understand why they have multiplied both fractions to make the denominators the same. In this scenario, the teacher should prompt the learner to try again or ask another learner to explain why the statement is true instead. They could ask the learner to think about what they already know about fractions to help them. This gives the

learners a chance to activate relevant prior knowledge and then develop their understanding by integrating this new knowledge into their long-term memory.

Sometimes, even after multiple prompts, learners might struggle to provide a self-explanation. In these cases, teachers should provide the learners with an explanation instead. Although it is much more beneficial when the learner explains themselves, the teacher giving an explanation is better than no explanation being given at all. The teacher providing an explanation is much more passive than the learner generating the explanation themselves and integrating it into their long-term memory; however, this still provides an opportunity for the learner to connect the new knowledge with some prior knowledge.

References

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2. Rittle-Johnson, B., Loehr, A. M., & Durkin, K. (2017). Promoting self-explanation to improve mathematics learning: A meta-analysis and instructional design principles. ZDM, 49(4), 599-611. https://doi.org/10.1007/s11858-017-0834-z